

Alice Gao

University of Washington

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Areas of interest

Human-computer interaction • UI Understanding • Computational Interfaces • Human-AI interaction • Data-driven design tools

Education

2023-present PHD in Computer Science & Engineering, **University of Washington**
2021-2023 MSc in Computer Science & Engineering, **University of Washington**
2017-2021 BS in Computer Science, **Princeton University**

Publications

* denotes equal contribution

CONFERENCE AND JOURNAL PAPERS

- 2026 **Alice Gao**, Andrew N. Meltzoff, Maarten Sap, Katharina Reinecke. Framing an AI with Values Reduces AI Reliance in AI-supported Writing Tasks. *ACM Conference on Fairness, Accountability, and Transparency (FAccT '26)*. Montreal, QC, Canada. June 2026.
- Nino Migineishvili, **Alice Gao**, Adinawa Adjabodjou, Dhanaraj Thaku, René Just, Katharina Reinecke. Regulating AI: Where U.S. State Policy and HCI (Mis)align. *ACM Conference on Human Factors in Computing Systems (CHI '26)*. Barcelona, Spain. April 2026. doi.org/10.1145/3772318.3791424.
- 2025 **Alice Gao**,* Samyukta Jayakumar,* Marcello Maniglia,* Brian Curless, Ira Kemelmacher-Shlizerman, Aaron R. Seitz and Steven M. Seitz. Don't Look at the Camera: Achieving Perceived Eye Contact. *Journal of Vision* 2025; 25(11):8, doi.org/10.1167/jov.24.10.1094.
- 2024 **Alice Gao**, Wataru Akahori, Naomi Yamashita, and Katharina Reinecke. Using Slack in the US and Japan: Surfacing Cultural Asymmetries in Overcoming Shortcomings. *In Submission*.

WORKSHOP PAPERS & POSTERS

- 2022 K.J. Kevin Feng*, **Alice Gao***, Johanna Suvi Karrass*. Towards Semantically Aware Word Cloud Shape Generation. *Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)*. Bend, OR. October 2022.

PREPRINTS

- 2026 **Alice Gao**, Weixi Tong, Rishab Vempati, Katharina Reinecke, R. Benjamin Shapiro, Tianyi Zhang, Jason Wu. Training Computer Use Agents to Assess the Usability of Graphical User Interfaces. April 2026. *In Submission*. [arXiv preprint](#).
- Donghoon Shin, **Alice Gao**, Rock Pang, Katharina Reinecke, Emily Tseng. Interrogating Design Homogenization in Web Vibe Coding. March 2026. *In Submission*. [arXiv preprint](#).

Research Experience

- Jan 2024 - Present **Research Assistant, Wildlab, University of Washington**
Advisors: Katharina Reinecke and R. Benjamin Shapiro
- Developing novel machine-learning algorithm to train computer use agents (CUAs) to automatically perform automated usability testing.
 - Creating data-driven, machine learning models to learn user preferences of different demographics and automate generate of better, personalized UI designs.
 - Designed a new interventions to reduce AI overreliance during AI-supported writing by up to 30%.
- Sep 2021 - Jan 2024 **Research Assistant, GRAIL, University of Washington**
Advisors: Steve Seitz, Brian Curless, Ira Kemelmacher-Shlizerman
- Developed machine learning techniques to create a model to redirect attention and eye gaze for a more attentive video conferencing experience.
- Sep 2019 - Feb 2021 **Undergraduate Research Assistant, Princeton Vision & Learning Lab, Princeton University**
Advisor: Jia Deng; mentored by Hei Law
- Improved associative embedding predictions and grouping from a neural network to provide faster and more accurate predictions for a one-stage object detector in instance segmentation.

Teaching

- University of Washington (graduate)**
- Spring 2026 CSE446: Machine Learning, *Teaching Assistant*
- Fall 2025 CSE373: Data Structures & Algorithms, *Teaching Assistant*
- Spring 2024 CSE581: Computer Ethics, *Teaching Assistant*
- Fall 2024 CSEP557: Current Trends In Computer Graphics, *Teaching Assistant*
- Winter 2024 CSE581: Computer Ethics, *Teaching Assistant*
- Princeton University (undergraduate)**
- Spring 2020 COS426: Computer Graphics, *Undergraduate Teaching Assistant*
- Fall 2018-Fall 2019 COS226: Algorithms & Data Structures, *Course Grader*

Industry Experience

- Jun-Aug 2020 *Software Engineering Intern in Online Account Opening; Capital One; New York City, New York*
- Created new endpoint and unit tests, laid framework for streaming data platform, and launched new security groups.

- Jun-Aug 2019 *AI Design Lab Intern; Tezign; Shanghai, China*
Assisted in optimizing image retrieval to match similar designs and contributed to training a model in judging strong vs weak graphic designs.
- Jun-Aug 2018 *Technical and App Development Intern; Princeton Satellite Systems; Princeton, New Jersey*
Devised beginnings of gameplay for spacecraft simulation iOS app. Created 3D models for Phase II version of Direct Fusion Drive (DFD), a direct-drive, fusion-powered rocket engine.

Service and Leadership

- May 2018 -
Mar 2020 *co-President, Princeton Chinese Students Association*
Started a new Red Envelope fundraiser, acted as a liaison between guest speakers and CSA, participated in creating the first Asian+Students Council, organized and raised funding to host Steven Lim as a guest speaker, started weekly community nights (mahjong nights) that persist until today.

Skills

Languages: Python, R, JavaScript, SQL, TypeScript, C, Go

Frameworks: PyTorch, scikit-learn, Pandas, React, Sveltekit, Node.js, Django/DRF, Flask

Tools: Docker, Apptainer, Figma

Other: Machine Learning, Quantitative and qualitative research methods, Statistical modeling, Experimental design

Last updated: May 12, 2026